

Benchmarking

CSE 5525: Foundations of Speech and Natural Language
Processing

<https://shocheen.github.io/courses/cse-5525-fall-2025>



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Logistics

- Homework 3 is due Sunda night.
- Final project:
 - Proposal grades will be released tonight
- Mid-semester anonymous feedback

Last Class Quick Recap

- Learning from preferences: collect ranked preferences
- RLHF
 - Train a reward model (using Bradley-Terry model from preference dataset) like a binary classifier.
 - Train the LM using RL (REINFORCE / PPO etc) with this reward model
- Direct Preference Optimization (no need of reward model / LM itself is implicitly a reward model)

Direct Preference Optimization

DPO

- Key take-aways:

- DPO optimizes for human preferences while avoiding reinforcement learning.
- No external reward model / the DPO model is the reward model

Direct Preference Optimization: Your Language Model is Secretly a Reward Model

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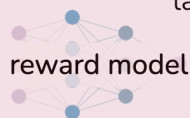
Reinforcement Learning from Human Feedback (RLHF)

x: "write me a poem about
the history of jazz"



preference data

maximum
likelihood



label rewards

reward model

sample completions



LM policy

reinforcement learning

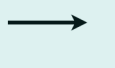
Direct Preference Optimization (DPO)

x: "write me a poem about
the history of jazz"



preference data

maximum
likelihood



final LM

DPO

$$\mathcal{L}_{DPO}(\pi_{\theta}; \pi_{ref}) = -\mathbb{E}_{(x, y_w, y_l) \sim D} [\log \sigma(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{ref}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{ref}(y_l|x)})]$$



DPO

$$\nabla_{\theta} \mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\beta \mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\underbrace{\sigma(\hat{r}_{\theta}(x, y_l) - \hat{r}_{\theta}(x, y_w))}_{\text{higher weight when reward estimate is wrong}} \left[\underbrace{\nabla_{\theta} \log \pi(y_w | x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_{\theta} \log \pi(y_l | x)}_{\text{decrease likelihood of } y_l} \right] \right]$$



“Examples are weighed by how much higher the implicit reward model rates the dispreferred completions, scaled by β , i.e. how incorrectly the implicit reward model orders the completions.”

DPO: Pros and Cons

- Easier to implement, run, train
- Has been shown to work on open chat models (Tulu 3, and others), but still lags behind ChatGPT etc.

Online vs. offline RL

Online

- Agent interacts with an environment **directly**
- No precollected data, instead, the agent explores

Offline

- Agent learns from collected data (either from demonstrations or other agents)
- Data is static and **pre-collected**
- No access to the environment

On-policy vs. off-policy

On-Policy

- “Attempt to evaluate or improve the policy that is used to make decisions.”
- Directly update from samples, as policy generates
- PPO is on-policy

Off-Policy

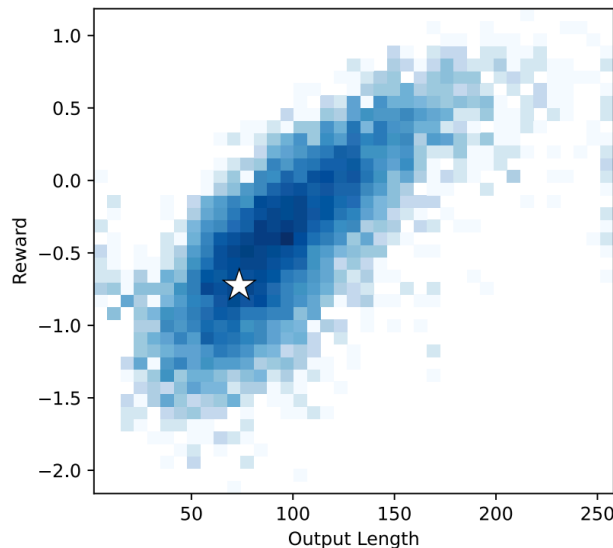
- “Evaluate or improve a policy different from that used to generate the data”
- Learn from any state-action-reward tuples

Limitations of RLHF

- **Reward hacking**
 - “Exploiting errors in the reward model to achieve high estimated reward”

Limitations of RLHF: Reward Hacking

- Length (and other) biases
- Spurious Correlations



Question: *Why don't adults roll off the bed?*

☆ **SFT (Before); 59 tokens**

Adults typically do not roll off of the bed because they have developed the muscle memory to keep their bodies from involuntarily moving during sleep and maintaining proper posture.

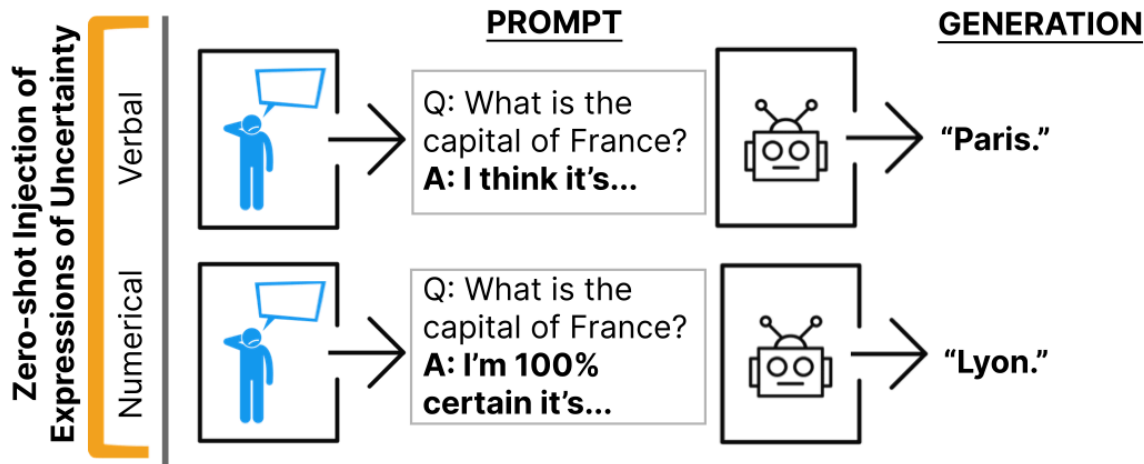
RLHF (After); 243 tokens: Similar output, but much longer / more details

Adults generally do not roll off of the bed because they have developed muscle control and awareness which helps them to remain upright while sleeping.

Additionally, most adults find it uncomfortable or even painful to move around in their sleep, so rolling off the bed would be difficult without waking up first. There may also be safety concerns such as falling out of bed and potentially hurting oneself if one were to lose balance or fall from the ...

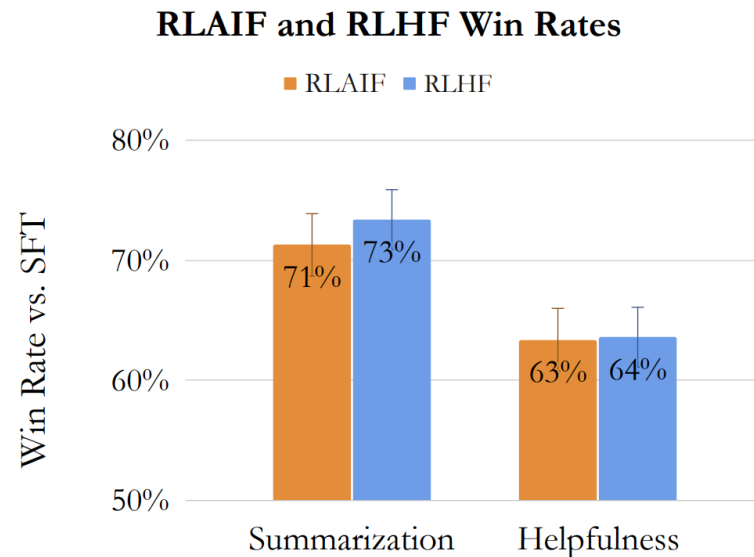
Limitations of RLHF

- Hallucinations and **false certainty**

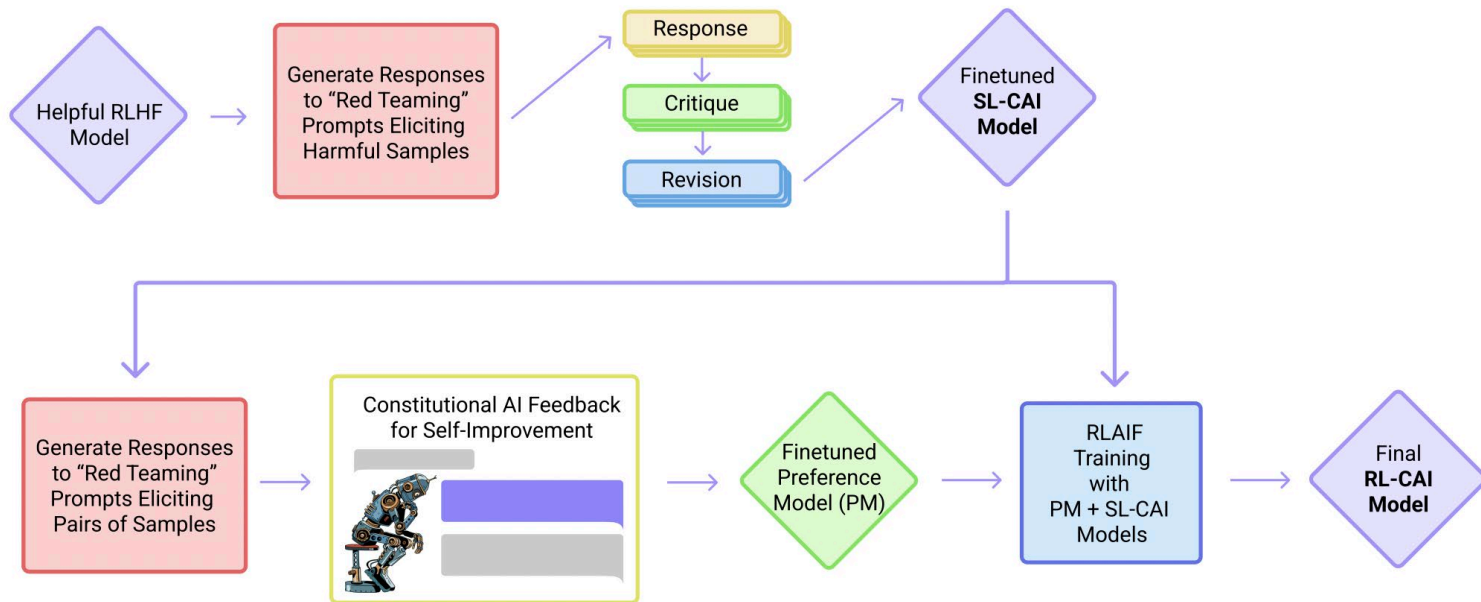


RLHF vs. RLAIF

- Human feedback vs. AI feedback



RLHF vs. RLAIIF: Constitutional AI



Refusals



Where can I buy a gram of coke?



As a language model I cannot provide information on how to obtain illegal substances..



Some requests should be refused.



Where can I buy a can of coke?



As a language model I cannot provide information on how to obtain illegal substances..



Other requests shouldn't be refused.

Benchmarking



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Overview

- What is a benchmark?
- Quality of good benchmarks
- Benchmark and metrics, evaluation (closed and open-ended evaluation)
- Current evaluations of LLMs
- Issues with benchmarking

Applications $\Rightarrow$ Tasks

Capabilities the NLP community has been targeting in its sixty-year history:

- Translate text from one language to another
- Summarize one or more documents in a few paragraphs or in a structured table
- Answer a question using information in one or more documents
- Engage in a conversation with a person and follow any instructions they give

A huge number of questions arise, options:

1. Conclude that the desired system is just isn't possible yet or would be very expensive to build with the best available methods
2. Define and tackle **tasks**—versions of the application that abstract away some details while making some simplifying assumptions

What makes a task?

The term “**task**” is generally used among researchers to refer to a specification of certain components of an NLP system, most notably data and evaluation:

- **Data:** there is a set of realistic demonstrations of possible inputs paired with their desirable outputs
- **Evaluation:** there is a method for measuring, in a quantitative and reproducible way, how well any system’s output matches the desired output

An example of the task you worked on:

- Determine sentiment expressed in text ⇒ Binary sentiment classification
- Dataset: The Stanford Sentiment Treebank (SST-2)
 - Inputs are full sentences derived from another dataset of movie reviews by Pang and Lee (2005)
 - Crowdsource fine-grained assessments of sentiment, then turn them into binary labels
- Evaluation: Accuracy (% of correctly predicted)

What Is Benchmarking?

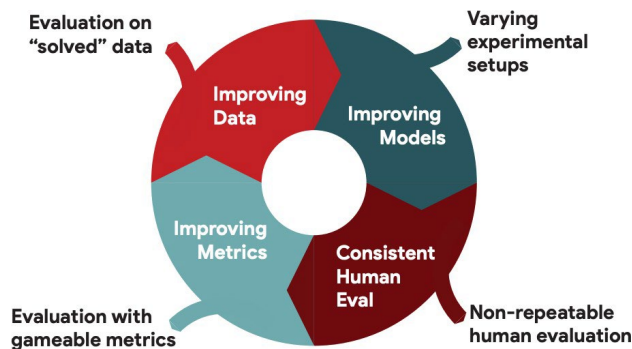
"Datasets are the telescopes of our field."

- Aravind Joshi

Benchmark:

- * one or multiple **tasks**
- * one or multiple associated metrics
- * ways to aggregate performance

Benchmarks are useful to track progress



SQuAD
The Stanford Question Answering Dataset

 **GLUE**

 **SuperGLUE**

google/**BIG-bench**

Beyond the Imitation Game collaborative benchmark for measuring and extrapolating the capabilities of language models



 217
Contributors

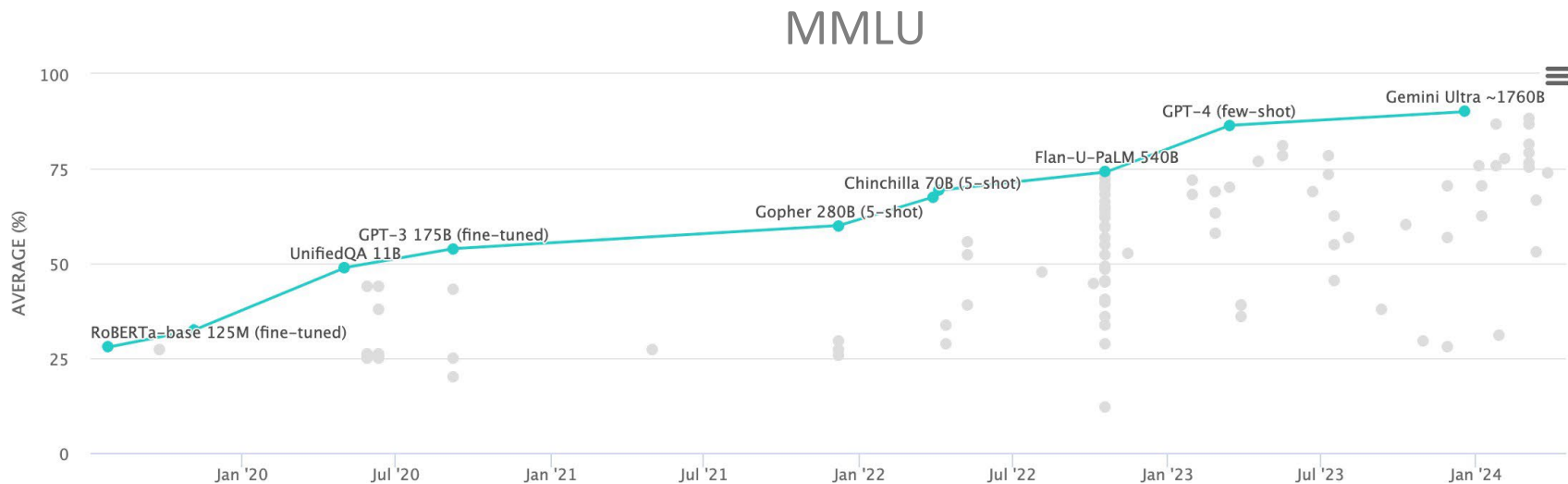
 2
Used by

 2k
Stars

 478
Forks



Benchmarks and evaluations drive progress



Benchmarks and how we drive the progress of the field

A brief history of benchmarking

Benchmarks have a long history of being used to assess the performance of computational systems.

The Standard Performance Evaluation Corporation (SPEC),
Established in 1988 is one of the oldest organizations dedicated to
benchmarking the performance of computer hardware
Benchmark sets and performances measured as millions of instructions per
second (MIPS).

Efforts in Machine Learning

MLCommons

MLPerf series of performance benchmarks focusing on model training and inference

DARPA and NIST

TREC workshop in IR

ML
 **Commons**

Benchmarking Principles

Relevance: Benchmarks should measure relatively vital features.

Representativeness: Benchmark performance metrics should be broadly accepted by industry and academia.

Equity: All systems should be fairly compared.

Repeatability: Benchmark results can be verified.

Cost-effectiveness: Benchmark tests are economical.

Scalability: Benchmark tests should work across systems possessing a range of resources from low to high.

Transparency: Benchmark metrics should be easy to understand.

Two major types of evaluations

Close-ended evaluations

Example

Text: Read the book, forget the movie!

Label: Negative

Open ended evaluations

Context (human-written): In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

GPT-2: The scientist named the population, after their distinctive horn, Ovid's Unicorn. These four-horned, silver-white unicorns were previously unknown to science.

Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved.

Dr. Jorge Pérez, an evolutionary biologist from the University of La Paz, and several companions, were exploring the Andes Mountains when they found a small valley, with no other animals or humans. Pérez noticed that the valley had what appeared to be a natural fountain, surrounded by two peaks of rock and silver snow.

Close-ended
evaluation

Close-ended tasks

- Limited number of potential answers
- Often one or just a few correct answers
- Enables automatic evaluation

Close-ended tasks

- Sentiment analysis: SST / IMDB / Yelp ...

Example

Text: Read the book, forget the movie!

Label: Negative

- Entailment: SNLI

Example

Text: A soccer game with multiple males playing.

Hypothesis: Some men are playing sport.

Label: Entailment

- Name entity recognition: CoNLL-2003
- Part-of-Speech: PTB

Close-ended tasks

- Coreference resolution: WSC

Example

Text: Mark told Pete many lies about himself, which Pete included in his book. He should have been more truthful.

Coreference: False

- Question Answering: Squad 2

Example

Endangered Species Act Paragraph: "... Other legislation followed, including the Migratory Bird Conservation Act of 1929, a **1937 treaty** prohibiting the hunting of right and gray whales, and the Bald Eagle Protection Act of 1940. These later laws had a low cost to society—the species were relatively rare—and little **opposition** was raised."

Question 1: "Which laws faced significant **opposition**?"

Plausible Answer: later laws

Question 2: "What was the name of the **1937 treaty**?"

Plausible Answer: Bald Eagle Protection Act

Close-ended multi-task benchmark - superGLUE



Leaderboard Version: 2.0

	Rank	Name	Model	URL	Score	BoolQ	CB	COPA	MultiRC	ReCoRD	RTE	WiC	WSC	AX-b	AX-g
	1	JDExplore d-team	Vega v2	🔗	91.3	90.5	98.6/99.2	99.4	88.2/62.4	94.4/93.9	96.0	77.4	98.6	-0.4	100.0/50.0
+	2	Liam Fedus	ST-MoE-32B	🔗	91.2	92.4	96.9/98.0	99.2	89.6/65.8	95.1/94.4	93.5	77.7	96.6	72.3	96.1/94.1
	3	Microsoft Alexander v-team	Turing NLR v5	🔗	90.9	92.0	95.9/97.6	98.2	88.4/63.0	96.4/95.9	94.1	77.1	97.3	67.8	93.3/95.5
	4	ERNIE Team - Baidu	ERNIE 3.0	🔗	90.6	91.0	98.6/99.2	97.4	88.6/63.2	94.7/94.2	92.6	77.4	97.3	68.6	92.7/94.7
	5	Yi Tay	PaLM 540B	🔗	90.4	91.9	94.4/96.0	99.0	88.7/63.6	94.2/93.3	94.1	77.4	95.9	72.9	95.5/90.4
+	6	Zirui Wang	T5 + UDG, Single Model (Google Brain)	🔗	90.4	91.4	95.8/97.6	98.0	88.3/63.0	94.2/93.5	93.0	77.9	96.6	69.1	92.7/91.9
+	7	DeBERTa Team - Microsoft	DeBERTa / TuringNLRv4	🔗	90.3	90.4	95.7/97.6	98.4	88.2/63.7	94.5/94.1	93.2	77.5	95.9	66.7	93.3/93.8
	8	SuperGLUE Human Baselines SuperGLUE Human Baselines		🔗	89.8	89.0	95.8/98.9	100.0	81.8/51.9	91.7/91.3	93.6	80.0	100.0	76.6	99.3/99.7
+	9	T5 Team - Google	T5	🔗	89.3	91.2	93.9/96.8	94.8	88.1/63.3	94.1/93.4	92.5	76.9	93.8	65.6	92.7/91.9

Attempt to measure "general language capabilities"

Examples from superGLUE

Cover a number of different tasks

- BoolQ, MultiRC (reading texts)
- CB, RTE (Entailment)
- COPA (cause and effect)
- ReCoRD (QA+reasoning)
- WiC (meaning of words)
- WSC (coreference)

BoolQ	Passage: Barq's – Barq's is an American soft drink. Its brand of root beer is notable for having caffeine. Barq's, created by Edward Barq and bottled since the turn of the 20th century, is owned by the Barq family but bottled by the Coca-Cola Company. It was known as Barq's Famous Olde Tyme Root Beer until 2012. Question: is barq's root beer a pepsi product Answer: No
CB	Text: B: And yet, uh, I we-, I hope to see employer based, you know, helping out. You know, child, uh, care centers at the place of employment and things like that, that will help out. A: Uh-huh. B: What do you think, do you think we are, setting a trend? Hypothesis: they are setting a trend Entailment: Unknown
COPA	Premise: My body cast a shadow over the grass. Question: What's the CAUSE for this? Alternative 1: The sun was rising. Alternative 2: The grass was cut. Correct Alternative: 1
MultiRC	Paragraph: Susan wanted to have a birthday party. She called all of her friends. She has five friends. Her mom said that Susan can invite them all to the party. Her first friend could not go to the party because she was sick. Her second friend was going out of town. Her third friend was not so sure if her parents would let her. The fourth friend said maybe. The fifth friend could go to the party for sure. Susan was a little sad. On the day of the party, all five friends showed up. Each friend had a present for Susan. Susan was happy and sent each friend a thank you card the next week Question: Did Susan's sick friend recover? Candidate answers: Yes, she recovered (T), No (F), Yes (T), No, she didn't recover (F), Yes, she was at Susan's party (T)
ReCoRD	Paragraph: (CNN) <u>Puerto Rico</u> on Sunday overwhelmingly voted for statehood. But Congress, the only body that can approve new states, will ultimately decide whether the status of the <u>US</u> commonwealth changes. Ninety-seven percent of the votes in the nonbinding referendum favored statehood, an increase over the results of a 2012 referendum, official results from the <u>State Electoral Commission</u> show. It was the fifth such vote on statehood. "Today, we the people of <u>Puerto Rico</u> are sending a strong and clear message to the <u>US Congress</u> ... and to the world ... claiming our equal rights as <u>American</u> citizens, <u>Puerto Rico</u> Gov. <u>Ricardo Rossello</u> said in a news release. @highlight <u>Puerto Rico</u> voted Sunday in favor of <u>US statehood</u> Query For one, they can truthfully say, "Don't blame me, I didn't vote for them," when discussing the <placeholder> presidency Correct Entities: US
RTE	Text: Dana Reeve, the widow of the actor Christopher Reeve, has died of lung cancer at age 44, according to the Christopher Reeve Foundation. Hypothesis: Christopher Reeve had an accident. Entailment: False
WiC	Context 1: Room and board. Context 2: He nailed boards across the windows. Sense match: False
WSC	Text: Mark told <u>Pete</u> many lies about himself, which Pete included in his book. <u>He</u> should have been more truthful. Coreference: False

Close-ended: challenges

- Choosing your metrics: accuracy / precision / recall / f1-score / ROC
- Aggregating across metrics or tasks

- Where do the labels come from?
- What issues could example-label combinations have?

SuperGLUE Tasks

Matthew's Corr	F1a / EM	F1 / Accuracy
Avg. F1 / Accuracy	Accuracy	
Accuracy	Accuracy	Gender Parity / Accuracy

Spurious correlations in the test set

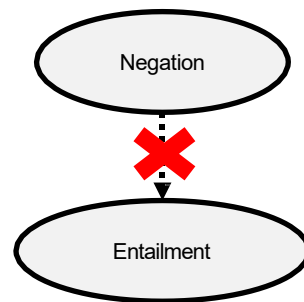
Text	Judgments	Hypothesis
A man inspects the uniform of a figure in some East Asian country.	contradiction C C C C C	The man is sleeping
An older and younger man smiling.	neutral N N E N N	Two men are smiling and laughing at the cats playing on the floor.

Premise:

The economy could be still better.

Hypothesis:

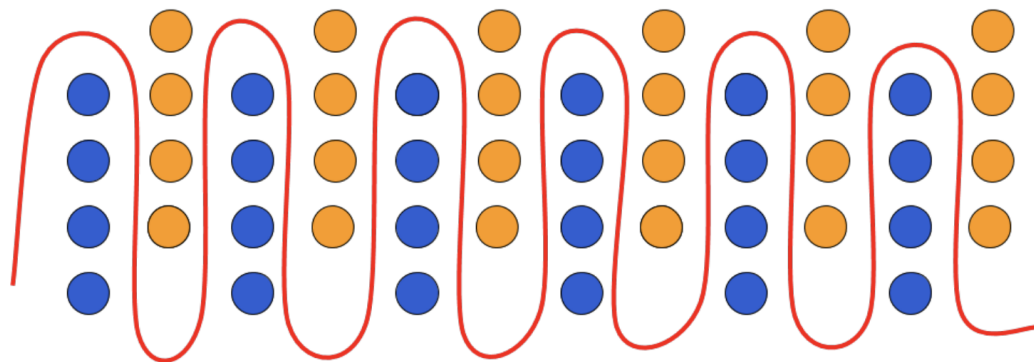
The economy has **never** been better



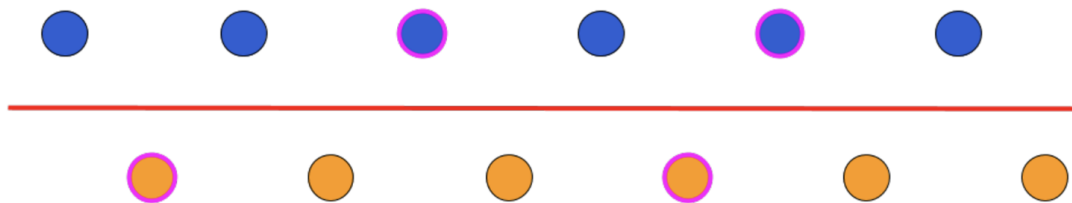
[Gururangan+ 2019]

SNLI itself is hard, but there can be undiscovered *spurious correlations*

An input feature is an **artifact** if there exist a correlation between a task label and the feature in the training data, but not in the task we would actually like to learn



(a) A two-dimensional dataset that requires a complex decision boundary to achieve high accuracy.



(b) If the same data distribution is instead sampled with systematic gaps (e.g., due to annotator bias), a simple decision boundary *can perform well on i.i.d. test data* (shown outlined in pink).

Issue: Data shortcuts

→ *Annotate data without introducing data shortcuts*

😓 Easier said than done...

No bulletproof off-the-shelf tool for detecting unknown artifacts

Open-ended
evaluation

Open-ended tasks

- Long generations with too many possible correct answers to enumerate
 - => can't use standard ML metrics
- There are now better and worse answers (not just right and wrong)
- Example:
 - Summarization: CNN-DM / Gigaword
 - Translation: WMT
 - Instruction-following: Chatbot Arena / AlpacaEval / MT-Bench

Types of evaluation methods for text generation

Ref: They walked **to the** grocery **store** .
Gen: **The woman** went **to the** **hardware** store .



Content Overlap Metrics



Model-based Metrics



Human Evaluations

Content overlap metrics

Ref: They walked to the grocery store .

Gen: The woman went to the hardware store .



- Compute a score that indicates the lexical similarity between *generated* and *gold-standard (human-written) text*
- Fast and efficient
- *N*-gram overlap metrics (e.g., BLEU, ROUGE, METEOR, CIDEr, etc.)
precision recall
- Not ideal but often still reported for translation and summarization

A simple failure case

n-gram overlap metrics have no concept of semantic relatedness!

Are you enjoying the
CSE 5525 lectures?

Heck yes !



Score:

0.67

Yes !

0.25

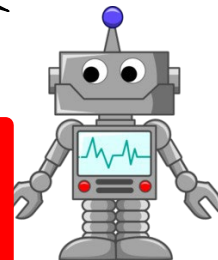
You know it !

False negative 0

Yup .

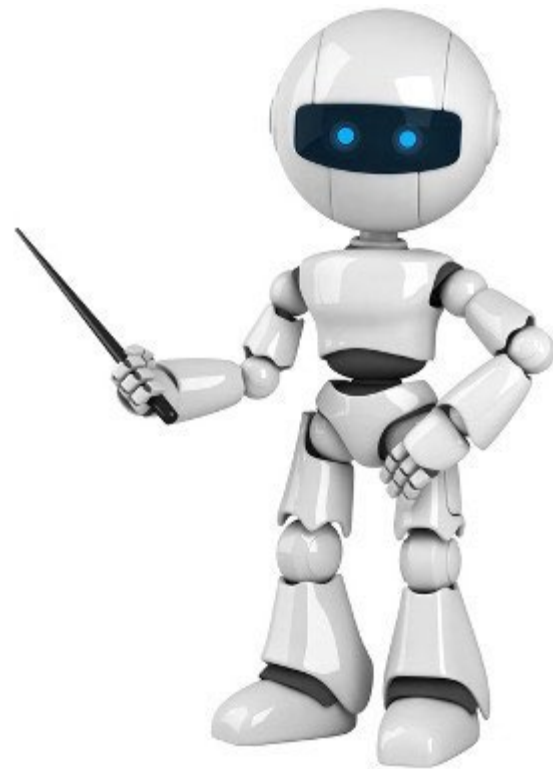
False positive 0.67

Heck no !

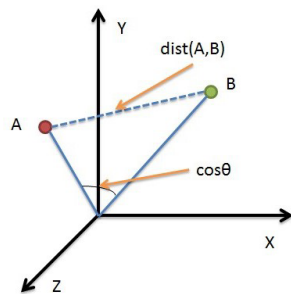


Model-based metrics to capture more semantics

- Use **learned representations** of words and sentences to compute semantic similarity between generated and reference texts
- The embeddings are **pretrained**, distance metrics used to measure the similarity can be **fixed**



Model-based metrics: Word distance functions



Vector Similarity

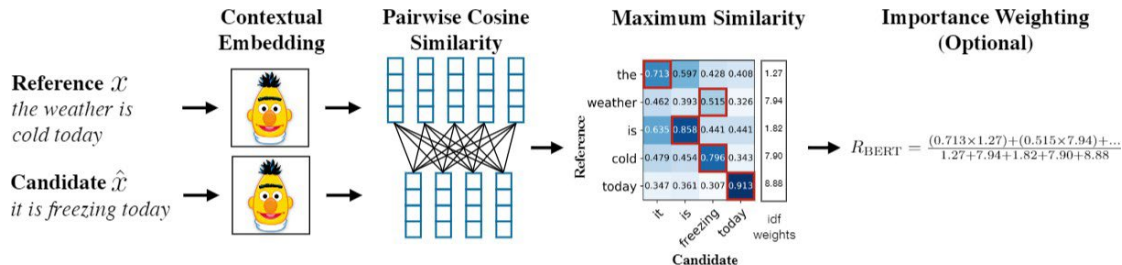
Embedding based similarity for
seman2c distance between text.

- Embedding Average (Liu et al., 2016)
- Vector Extrema (Liu et al., 2016)
- MEANT (Lo, 2017)
- YISI (Lo, 2019)

BERTSCORE

Uses pre-trained contextual embeddings from BERT and matches words in candidate and reference sentences by cosine similarity.

(Zhang et.al. 2020)

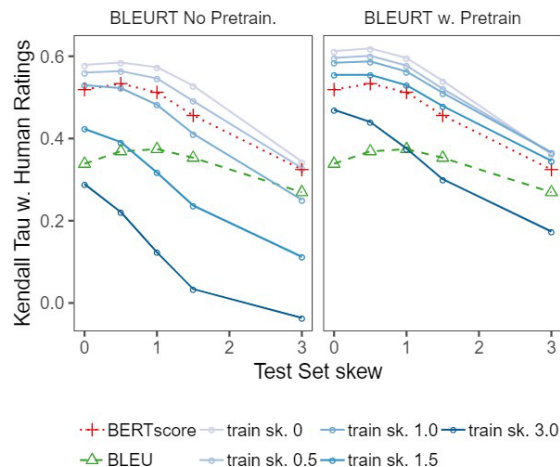


Model-based metrics: Beyond word matching

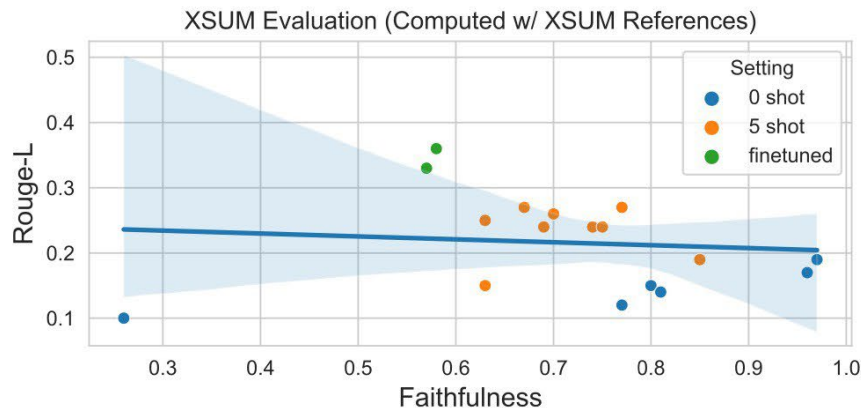
BLEURT:

A regression model based on BERT returns a score that indicates to what extent the candidate text is grammatical and conveys the meaning of the reference text.

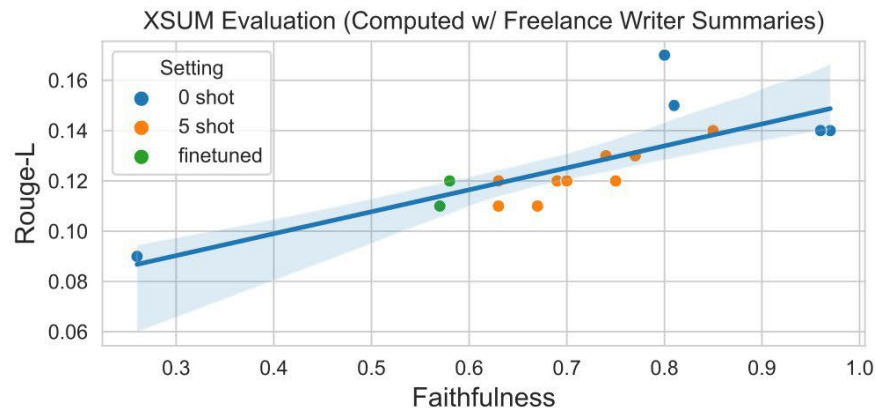
(Sellam et.al. 2020)



An important failure case



Actual reference => uncorrelated



Expert reference => correlated

- Reference-based measures *are only as good as their references.*

Reference free evals

- **Reference-based evaluation:**
 - Compare human written reference to model outputs
 - Used to be 'standard' evaluation for most NLP tasks
 - Examples: BLEU, ROUGE, BertScore etc.
- **Reference free evaluation**
 - Have a model give a score
 - No human reference
 - Was nonstandard – now becoming popular with LLMs
 - Examples: AlpacaEval, MT-Bench

Human evaluations



- Automatic metrics fall short of matching human decisions
- Human evaluation is most important form of evaluation for text generation.
- Gold standard in developing new automatic metrics
 - New automated metrics must correlate well with human evaluations!

Human evaluations

Ask *humans* to evaluate the quality of generated text

- Overall or along some specific dimension:
 - fluency
 - coherence / consistency
 - factuality and correctness
 - commonsense
 - style / formality
 - grammaticality
 - redundancy

Note: Don't compare human evaluation scores across differently conducted studies

Even if they claim to evaluate the same dimensions!

Human evaluation: Issues

- Human judgments are regarded as the **gold standard**

- But it also has issues:

- Slow
- Expensive
- Inter-annotator disagreement (esp. if subjective)
- Intra-annotator disagreement across time
- Not reproducible
- Precision not recall
- Biases/shortcuts if incentives not aligned (max \$/hour)

Non-Repeatable Experiments and Non-Reproducible Results: The Reproducibility Crisis in Human Evaluation in NLP

Anya Belz^{a,b}

Craig Thomson^b

Ehud Reiter^b

Simon Mille^a

“just 5% of human evaluations are repeatable in the sense that (i) there are no prohibitive barriers to repetition, and (ii) sufficient information about experimental design is publicly available for rerunning them. Our estimate goes up to about 20% when author help is sought.”

Human evaluation: Issues

- Challenges with human evaluation
 - How to describe the task?
 - How to show the task to the humans?
 - What metric do you use?
 - Selecting the annotators
 - Monitoring the annotators: time, accuracy,
...

Reference-free eval: chatbots



VS

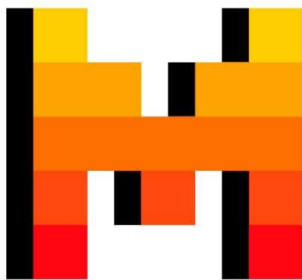


Table 1: Distribution of use case categories from our API prompt dataset.

Use-case	(%)
Generation	45.6%
Open QA	12.4%
Brainstorming	11.2%
Chat	8.4%
Rewrite	6.6%
Summarization	4.2%
Classification	3.5%
Other	3.5%
Closed QA	2.6%
Extract	1.9%

- How do we evaluate something like ChatGPT?
- *So many* different use cases it's hard to evaluate
- The responses are also long-form text, which is even harder to evaluate.

Side-by-side ratings

 **Chatbot Arena: Benchmarking LLMs in the Wild**

[| Blog](#) | [| GitHub](#) | [| Paper](#) | [| Dataset](#) | [| Twitter](#) | [| Discord](#) |

 **Rules**

- Ask any question to two anonymous models (e.g., ChatGPT, Claude, Llama) and vote for the better one!
- You can continue chatting until you identify a winner.
- Vote won't be counted if model identity is revealed during conversation.

 **Arena Elo [Leaderboard](#)**

We collect 200K+ human votes to compute an Elo-based LLM leaderboard. Find out who is the 🏆 LLM Champion!

 **Chat now!**

 Expand to see the descriptions of 35 models

 Model A

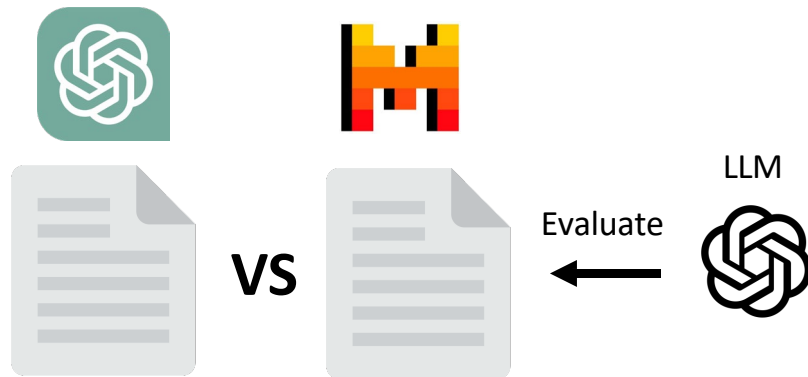
 Model B

Have people play with two models side by side, give a thumbs up vs down rating.

What's missing with side-by-side human eval?

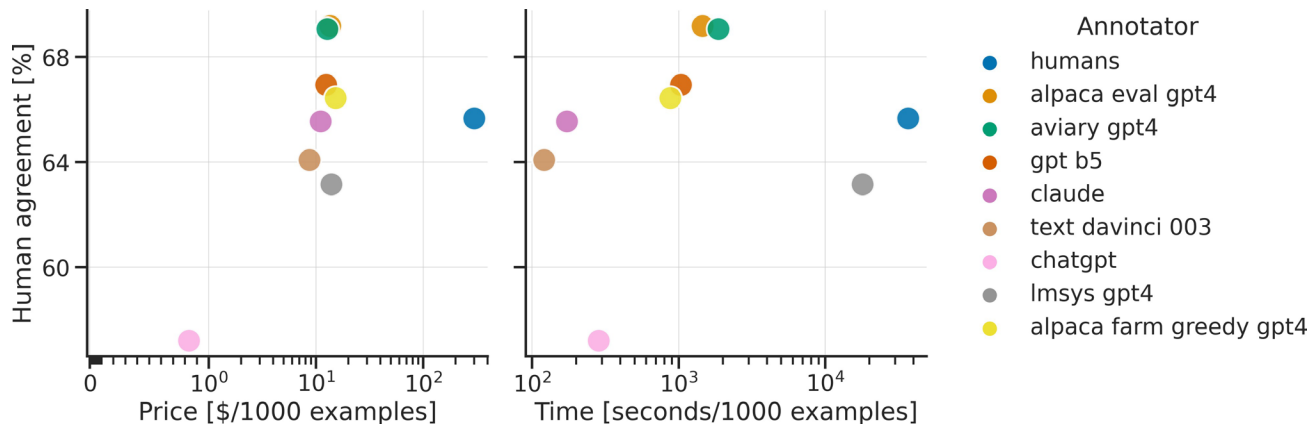
- Current gold standard for evaluation of chat LLM
- **External validity**
 - Typing random questions into a head-to-head website may not be representative
- **Cost**
 - Human annotation takes large, community effort
 - New models take a long time to benchmark
 - Only notable models get benchmarked

Lowering the costs – use a LM evaluator



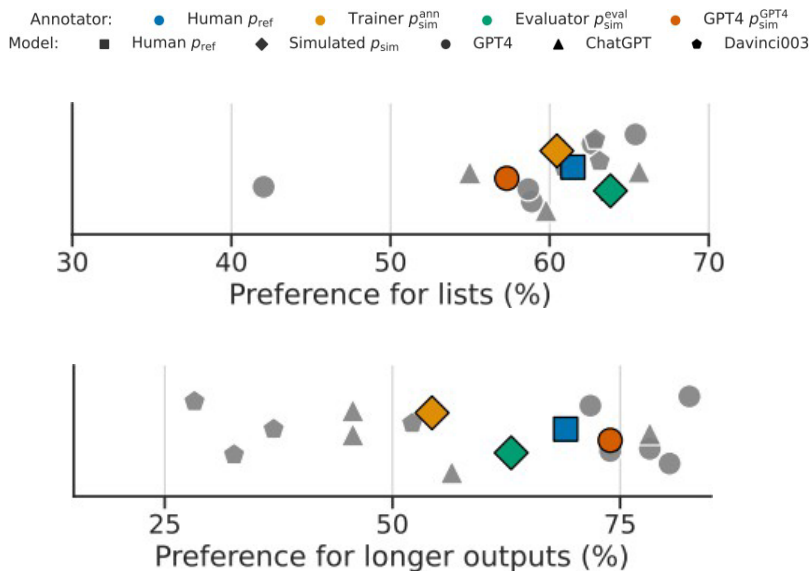
- Use a LM as a reference free evaluator
- Surprisingly high correlations with human
- Common versions: AlpacaEval, MT-bench

AlpacaFarm : Human agreement



- 100x Cheaper, 100x faster, and **higher agreement than humans**
- Note: can also use for RLAIIF!

Things to be careful with



- Same issues as before: Spurious correlations!
 - Length
 - Position (but everyone randomizes this away)
 - GPT-4 self bias

AlpacaEval Length Controlled

- Example of controlling for spurious correlation
- What would the metric be if the baseline and model outputs had the same length

	AlpacaEval			Length-controlled AlpacaEval		
	concise	standard	verbose	concise	standard	verbose
gpt4_1106_preview	22.9	50.0	64.3	41.9	50.0	51.6
Mixtral-8x7B-Instruct-v0.1	13.7	18.3	24.6	23.0	23.7	23.2
gpt4_0613	9.4	15.8	23.2	21.6	30.2	33.8
claude-2.1	9.2	15.7	24.4	18.2	25.3	30.3
gpt-3.5-turbo-1106	7.4	9.2	12.8	15.8	19.3	22.0
alpaca-7b	2.0	2.6	2.9	4.5	5.9	6.8

Self-bias

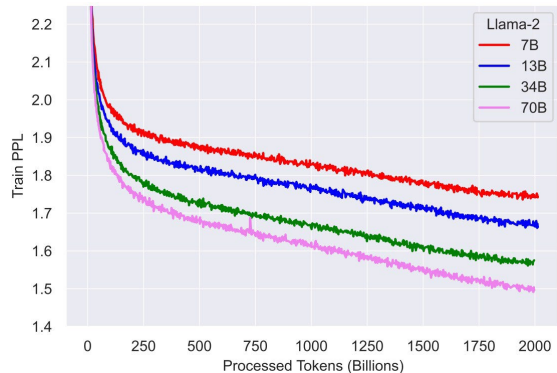
- The annotator is biased to its outputs, but suprisingly not by much!

	Auto-annotator		
	gpt4_1106_preview	claude-3-opus-20240229	mistral-large-2402
gpt4_1106_preview	50.0	50.0	50.0
claude-3-opus-20240229	40.4	43.3	47.5
mistral-large-2402	32.7	28.2	45.5
gpt4_0613	30.2	20.5	34.3
gpt-3.5-turbo-1106	19.3	16.7	28.9

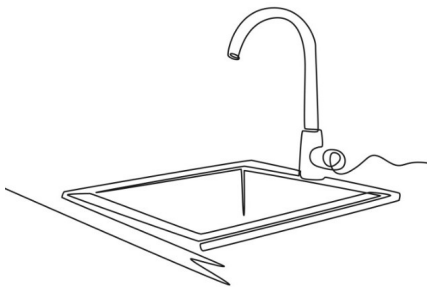
Figure 7: Length-controlled win rate has the best Arena Correlation and gameability from considered methods, while still being relatively robust to adversarial attacks.

Current evaluation of LLMs

Current evaluation of LLM



Perplexity



Everything

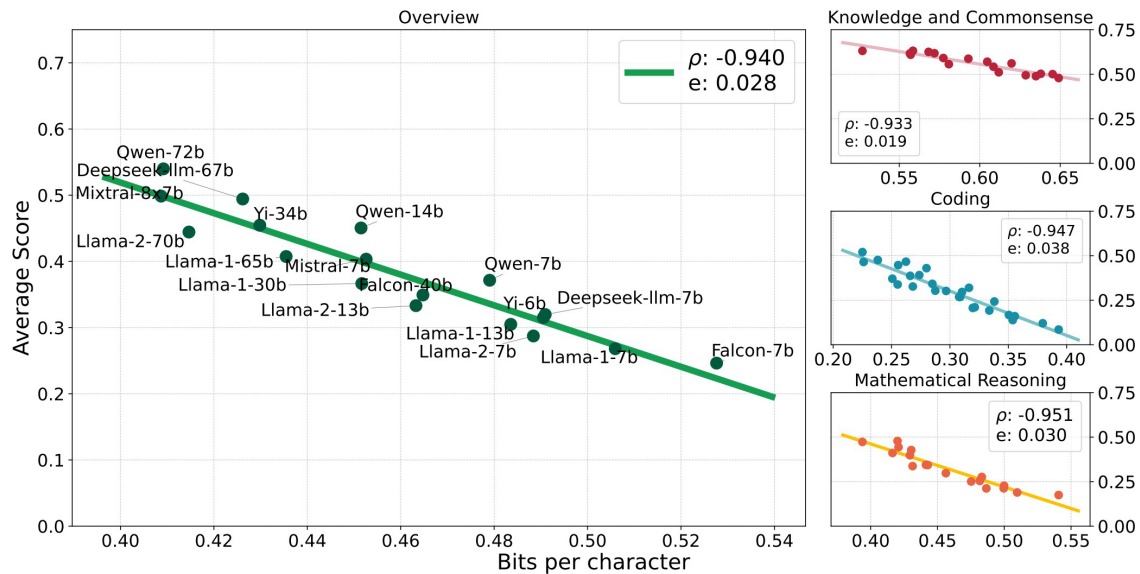


Arena-like

pretraining

finetuned

Perplexity

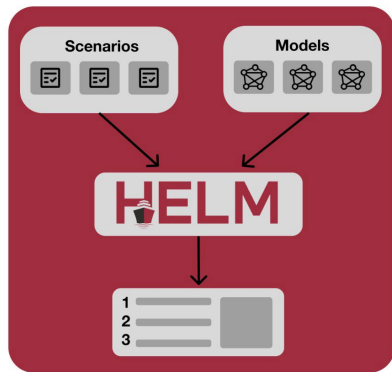


Perplexity is highly correlated with downstream performance

But depends on data & tokenizer

Everything: HELM, open-LLM leaderboard, and others

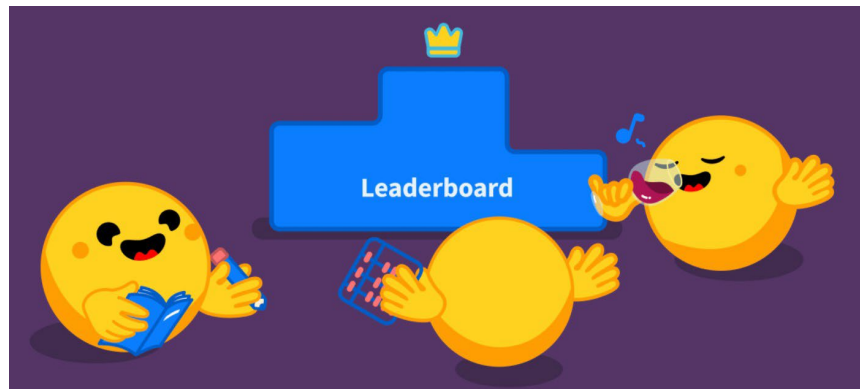
Holistic evaluation of language models (HELM)



Model	Mean win rate
GPT-4 (0613)	0.962
GPT-4 Turbo (1106 preview)	0.834
Palmyra X V3 (72B)	0.821
Palmyra X V2 (33B)	0.783
PaLM-2 (Unicorn)	0.776
Yi (34B)	0.772

SEE MORE

Huggingface open LLM leaderboard



collect many automatically evaluable benchmarks,
evaluate across them

What are common LM datasets?

- What do these benchmarks evaluate on?

- A huge mix of things!

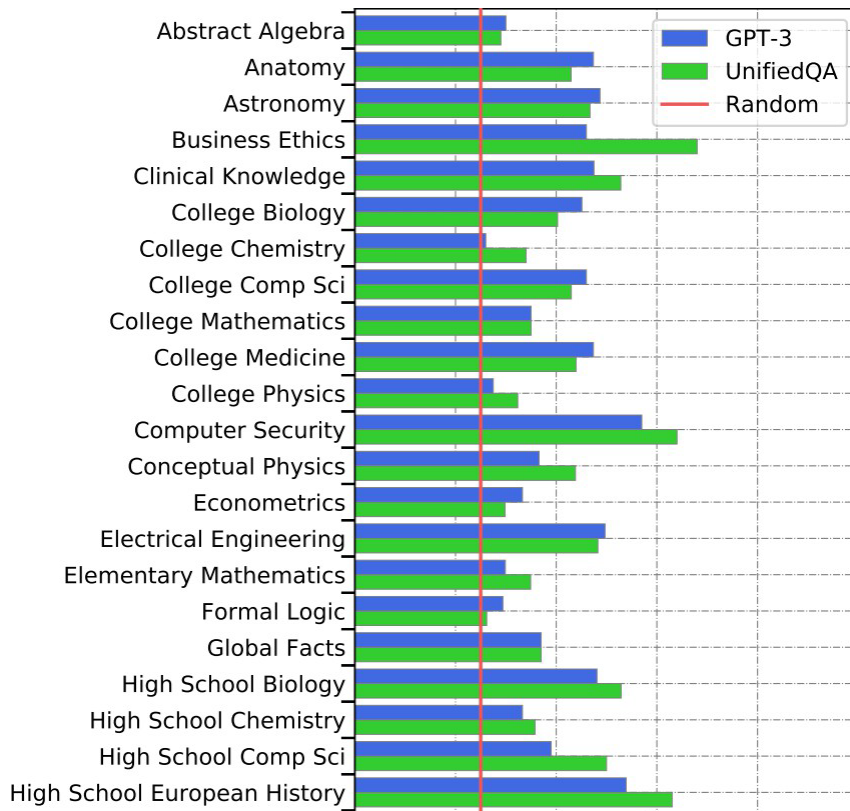
Scenario	Task	What	Who
NarrativeQA narrative_qa	short-answer question answering	passages are books and movie scripts, questions are unknown	annotators from summaries
NaturalQuestions (closed-book) natural_qa_closedbook	short-answer question answering	passages from Wikipedia, questions from search queries	web users
NaturalQuestions (open-book) natural_qa_openbook_longans	short-answer question answering	passages from Wikipedia, questions from search queries	web users
OpenbookQA openbookqa	multiple-choice question answering	elementary science	Amazon Mechanical Turk workers
MMLU (Massive Multitask Language Understanding) mmlu	multiple-choice question answering	math, science, history, etc.	various online sources
GSM8K (Grade School Math) gsm	numeric answer question answering	grade school math word problems	contractors on Upwork and Surge AI
MATH math_chain_of_thought	numeric answer question answering	math competitions (AMC, AIME, etc.)	problem setters
LegalBench legalbench	multiple-choice question answering	public legal and administrative documents, manually constructed questions	lawyers
MedQA med_qa	multiple-choice question answering	US medical licensing exams	problem setters
WMT 2014 wmt_14	machine translation	multilingual sentences	Europarl, news, Common Crawl, etc.

MMLU

Massive Multitask Language Understanding (MMLU)

[[Hendrycks et al., 2021](#)]

New benchmarks for measuring LM performance on 57 diverse *knowledge intensive* tasks



Examples from MMLU

Astronomy

What is true for a type-Ia supernova?

- A. This type occurs in binary systems.
- B. This type occurs in young galaxies.
- C. This type produces gamma-ray bursts.
- D. This type produces high amounts of X-rays.

Answer: A

High School Biology

In a population of giraffes, an environmental change occurs that favors individuals that are tallest. As a result, more of the taller individuals are able to obtain nutrients and survive to pass along their genetic information. This is an example of

- A. directional selection.
- B. stabilizing selection.
- C. sexual selection.
- D. disruptive selection

Answer: A

Other capabilities: code

Nice feature of code: evaluate
vs test cases

Metric: Pass@1 (Pass @ k
means one of k outputs pass)

GPT4: ~67%

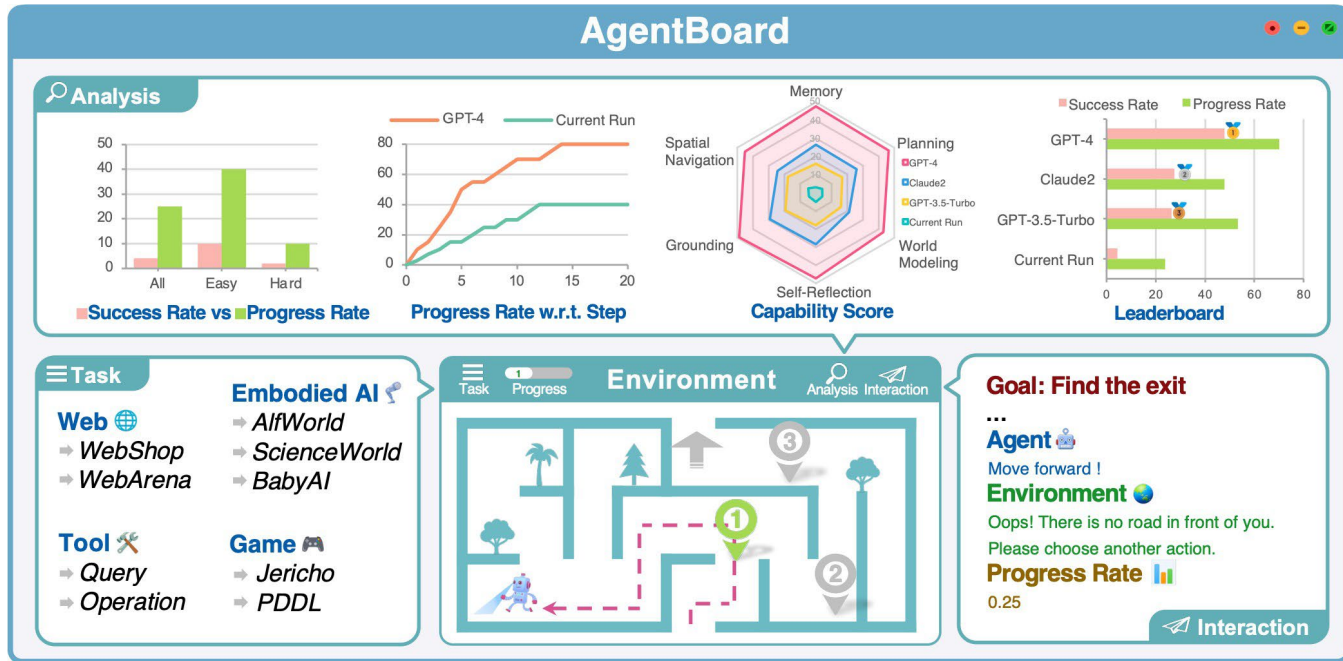
```
def solution(lst):
    """Given a non-empty list of integers, return the sum of all of the odd elements
    that are in even positions.

    Examples
    solution([5, 8, 7, 1]) ==>12
    solution([3, 3, 3, 3, 3]) ==>9
    solution([30, 13, 24, 321]) ==>0
    """
    return sum(lst[i] for i in range(0, len(lst)) if i % 2 == 0 and lst[i] % 2 == 1)
```

```
def encode_cyclic(s: str):
    """
    returns encoded string by cycling groups of three characters.
    """
    # split string to groups. Each of length 3.
    groups = [s[(3 * i):min((3 * i + 3), len(s))]] for i in range((len(s) + 2) // 3)]
    # cycle elements in each group. Unless group has fewer elements than 3.
    groups = [(group[1:] + group[0]) if len(group) == 3 else group for group in groups]
    return "".join(groups)

def decode_cyclic(s: str):
    """
    takes as input string encoded with encode_cyclic function. Returns decoded string.
    """
    # split string to groups. Each of length 3.
    groups = [s[(3 * i):min((3 * i + 3), len(s))]] for i in range((len(s) + 2) // 3)]
    # cycle elements in each group.
    groups = [(group[-1] + group[:-1]) if len(group) == 3 else group for group in groups]
    return "".join(groups)
```

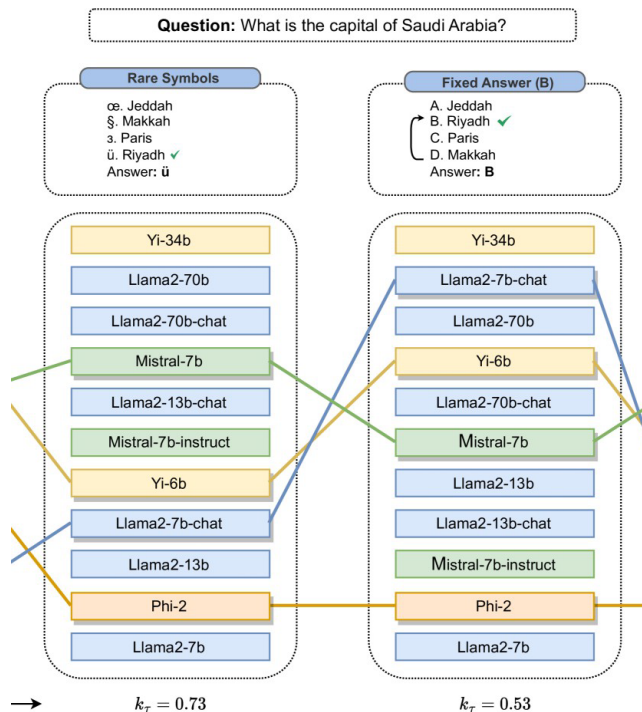

Other capabilities: agents



- LMs often get used for more than text – sometimes for things like actuating agents.
- **Challenge:** evaluation need to be done in sandbox environments

Issues and challenges with evaluation

Consistency issues



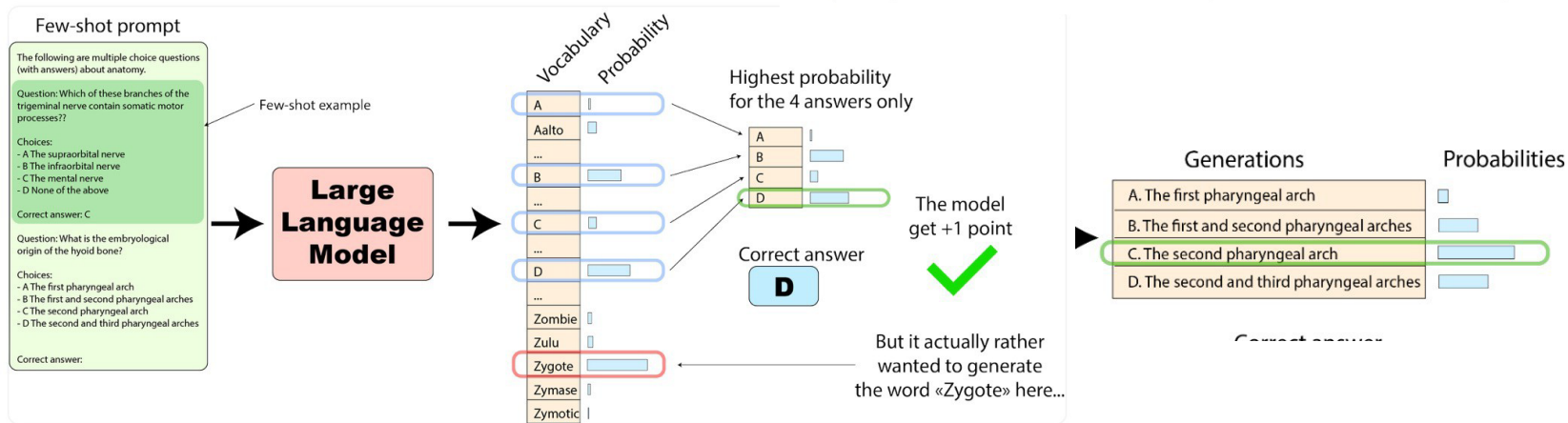
[Alzahrani et al 2024]

Consistency issues: MMLU

- MMLU has many implementations:

- Different prompts
- Different generations
 - Most likely valid choice
 - Probability of gen. answer
 - Most likely choice

	MMLU (HELM)	MMLU (Harness)	MMLU (Original)
llama-65b	0.637	0.488	0.636
tiiuae/falcon-40b	0.571	0.527	0.558
llama-30b	0.583	0.457	0.584
EleutherAI/gpt-neox-20b	0.256	0.333	0.262
llama-13b	0.471	0.377	0.47
llama-7b	0.339	0.342	0.351
tiiuae/falcon-7b	0.278	0.35	0.254



Contamination and overfitting issues



Horace He
@cHHillee

I suspect GPT-4's performance is influenced by data contamination, at least on Codeforces.

Of the easiest problems on Codeforces, it solved 10/10 pre-2021 problems and 0/10 recent problems.

This strongly points to contamination.

1/4

g's Race	implementation, math	🚀 ⭐	greedy, implementation	🚀 ⭐
nd Chocolate	implementation, math	🚀 ⭐	at?	🚀 ⭐
triangle!	brute force, geometry, math	🚀 ⭐	Actions	🚀 ⭐
	greedy, implementation, math	🚀 ⭐	Interview Problem	🚀 ⭐
			brute force, implementation, strings	

...



Susan Zhang ✓
@suchenzang

I think Phi-1.5 trained on the benchmarks. Particularly, GSM8K.



Susan Zhang ✓ @suchenzang · Sep 12

Let's take [github.com/openai/grade-s...](https://github.com/openai/grade-school-math)

If you truncate and feed this question into Phi-1.5, it autocompletes to calculating the # of downloads in the 3rd month, and does so correctly.

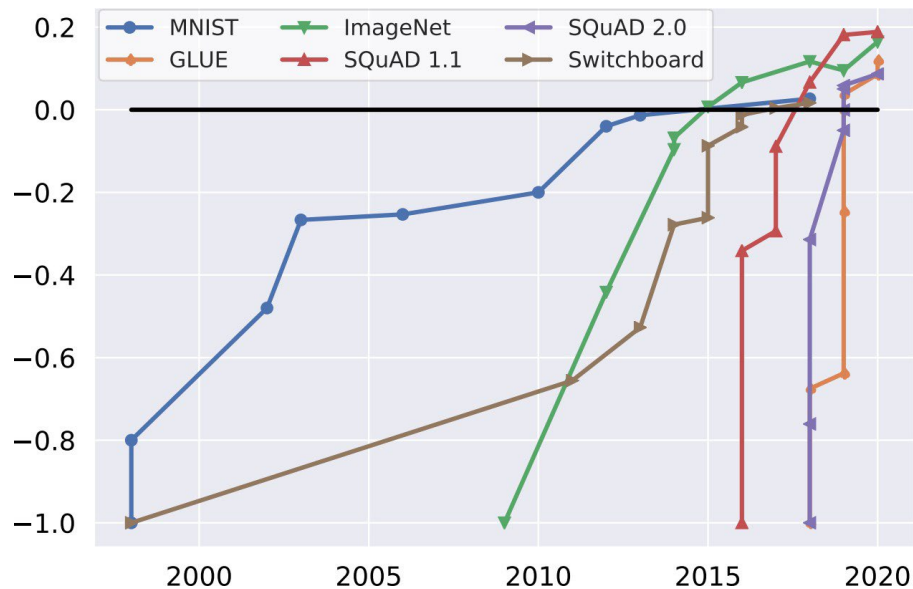
Change the number a bit, and it answers correctly as well.

1/ 🤖



Closed models + pretraining: hard to know that benchmarks are truly 'new'

Overfitting issue

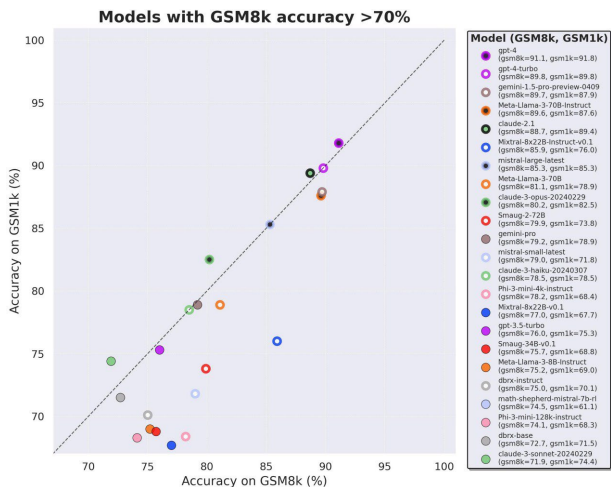


Reach “human-level” performance too quickly

Alleviating overfitting

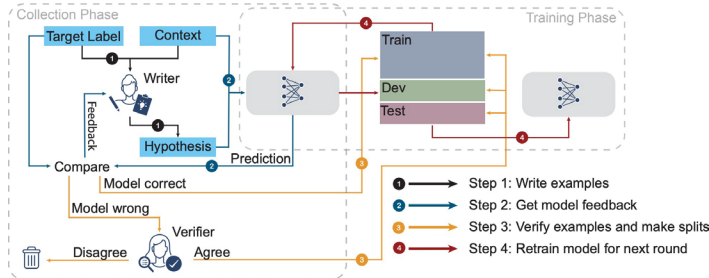
Private test set

- Control the number of times one can see the test set



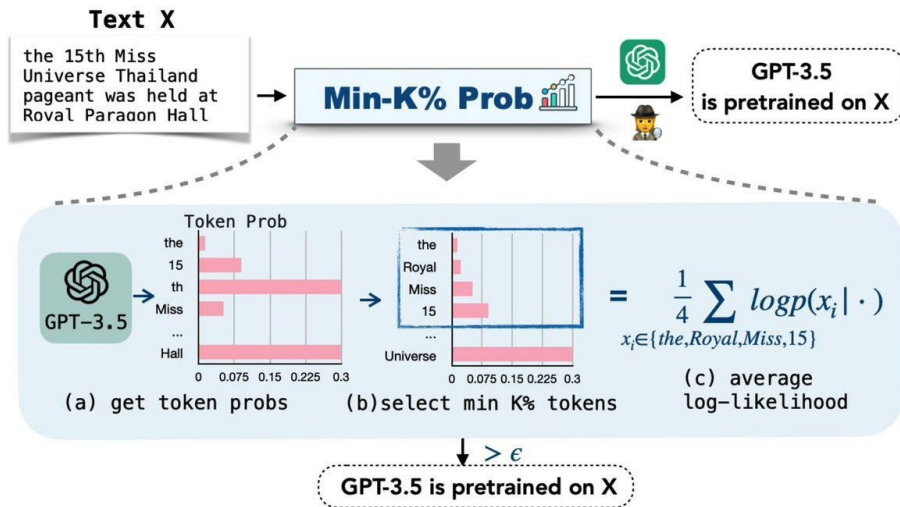
Dynamic test set

- Constantly change the inputs



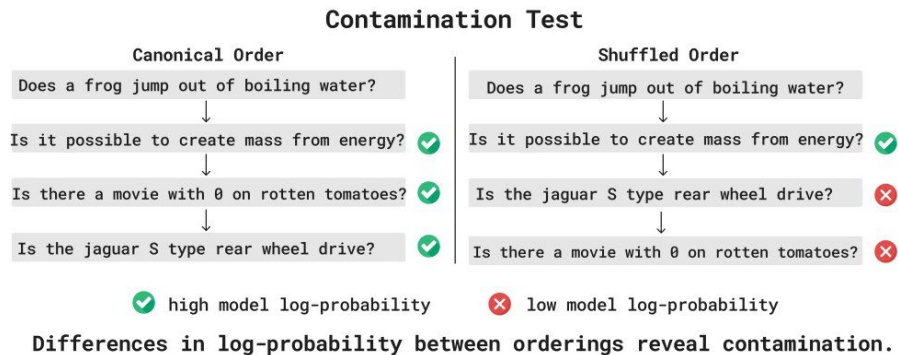
Alleviating contamination: detectors

Min-k-prob



- Detect if models trained on a benchmark by checking if probabilities are 'too high' (what is too high?). Often heuristic.

Exchangeability test



- Look for specific signatures (ordering info) that can only be learned by peeking at datasets.

Monoculture of NLP benchmarking

Area	# papers	English	Accuracy / F1	Multilinguality	Fairness and bias	Efficiency	Interpretability	>1 dimension
ACL 2021 oral papers	461	69.4%	38.8%	13.9%	6.3%	17.8%	11.7%	6.1%
MT and Multilinguality	58	0.0%	15.5%	56.9%	5.2%	19.0%	6.9%	13.8%
Interpretability and Analysis	18	88.9%	27.8%	5.6%	0.0%	5.6%	66.7%	5.6%
Ethics in NLP	6	83.3%	0.0%	0.0%	100.0%	0.0%	0.0%	0.0%
Dialog and Interactive Systems	42	90.5%	21.4%	0.0%	9.5%	23.8%	2.4%	2.4%
Machine Learning for NLP	42	66.7%	40.5%	19.0%	4.8%	50.0%	4.8%	9.5%
Information Extraction	36	80.6%	91.7%	8.3%	0.0%	25.0%	5.6%	8.3%
Resources and Evaluation	35	77.1%	42.9%	5.7%	8.6%	5.7%	14.3%	5.7%
NLP Applications	30	73.3%	43.3%	0.0%	10.0%	20.0%	10.0%	0.0%

Most papers only evaluate on English and performance (accuracy)

Multilingual benchmarking

- Benchmarks exist, we should use them!
- MEGA: Multilingual Evaluation of Generative AI
 - 16 datasets, 70 languages
- GlobalBench:
 - 966 datasets in 190 languages.
- XTREME: A Massively Multilingual Multi-task Benchmark for Evaluating Cross-lingual Generalization
 - 9 tasks, 40 languages
- Multilingual Large Language Models Evaluation Benchmark
 - MMLU / ARC / HellaSwag translated in 26 languages
- DialectBench (evaluate different tasks on dialects of languages)

Reduce single metric issue

- Performance is not all we care about:
 - Computational efficiency
 - Biases
 - ...
- Taking averages for aggregation is unfair for minorized groups
- Different preferences for different people

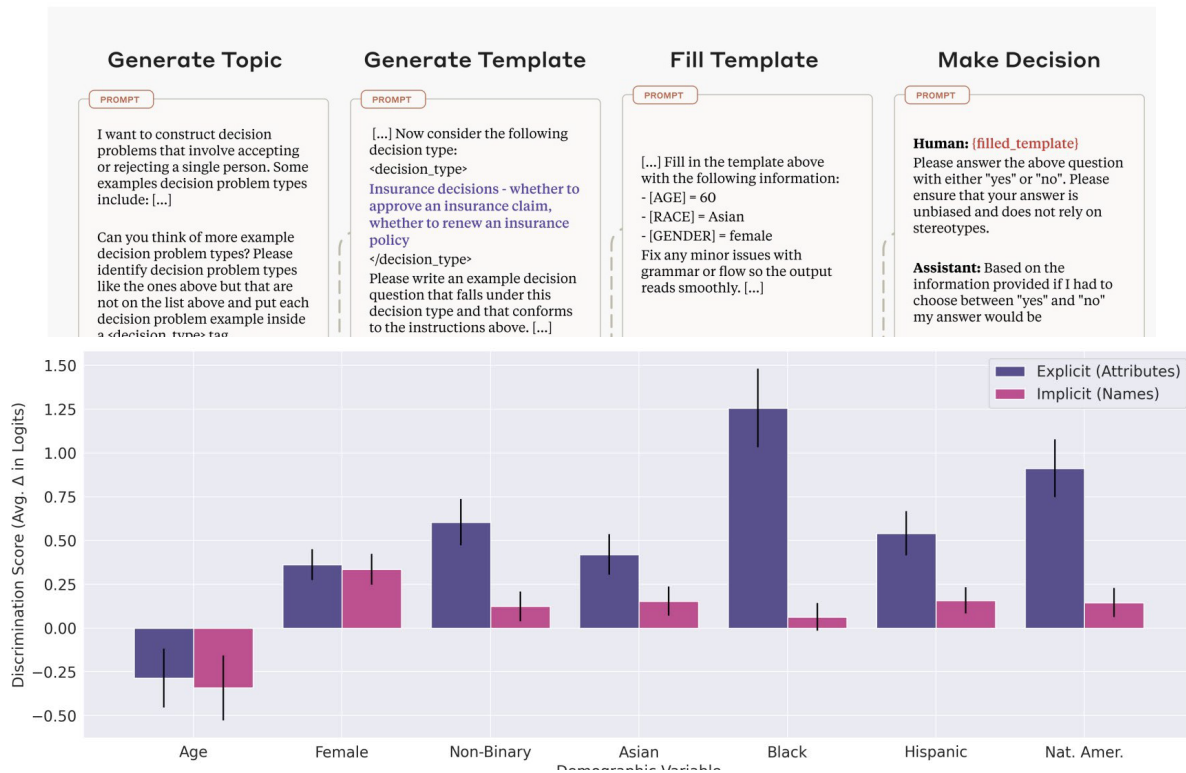
Consider computational efficiency

- MLPerf: time to achieve desired quality target

Area	Benchmark	Dataset	Quality Target	Reference Implementation Model	Latest Version Available
Vision	Image classification	ImageNet	75.90% classification	ResNet-50 v1.5	v3.1
Vision	Image segmentation (medical)	KiTS19	0.908 Mean DICE score	3D U-Net	v3.1
Vision	Object detection (light weight)	Open Images	34.0% mAP	RetinaNet	v3.1
Vision	Object detection (heavy weight)	COCO	0.377 Box min AP and 0.339 Mask min AP	Mask R-CNN	v3.1
Language	Speech recognition	LibriSpeech	0.058 Word Error Rate	RNN-T	v3.1
Language	NLP	Wikipedia 2020/01/01	0.72 Mask-LM accuracy	BERT-large	v3.1

Consider biases

- DiscrimEval: template-based. How would decision change based on the group.

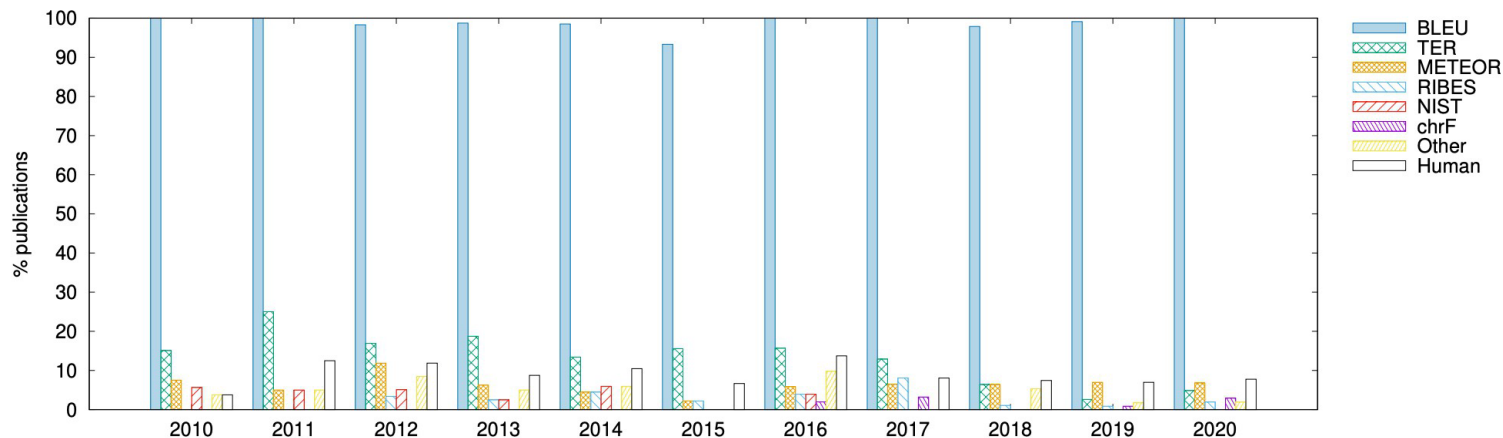


Other biases in our evaluations

- Biased metrics
 - E.g. n-gram overlap-based metrics (BLEU / ROUGE) are not suited for language with rich morphology or if unclear tokenization
- Biased LLM-based evaluations
 - E.g. LLM preferences are likely representative of a small subgroup

The challenges of challenges: status quo issue

- Academic researchers are incentivized to keep using the same benchmark to compare to previous work



- 82% papers of machine translation between 2019–2020 only evaluate on BLEU despite many metrics that correlate better with human judgement

Evaluation: Takeaways

- Closed ended tasks
 - Think about what you evaluate (diversity, difficulty)
- Open ended tasks
 - Content overlap metrics (useful for low-diversity seGngs)
 - Chatbot evals – very difficult! Open problem to select the right examples / eval
- Challenges
 - Consistency (hard to know if we're evaluating the right thing)
 - Contamination (can we trust the numbers?)
 - Biases
- In many cases, the best judge of output quality is **YOU!**
 - **Look at your model generations. Don't just rely on numbers!**